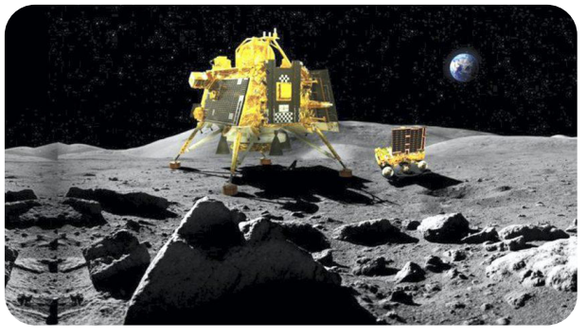


Figure 8.10 Lander of Chandrayaan-3

2023, and the Lander touched down successfully on August 23, 2023, on the surface of the Moon. Through this, India became the fourth country after Russia, America and China to land on the Moon successfully and the first to land on the lunar South Pole region.

Chandrayaan-3 spacecraft consists of Lander (Vikram) and Rover (Pragyan). The Lander can safely land and roll out the Rover on the lunar surface. The Lander and the Rover have scientific payloads to conduct studies on the geology and minerals present on the lunar surface.

The objectives of Chandrayaan-3 mission are:

- To safely land the Lander on the lunar surface.
- To make the Rover roam on the Moon.
- To conduct scientific experiments on the lunar surface.

Chandrayaan-3 mission has achieved many things. The Lander investigated the temperature of the lunar soil at various depths and the moonquakes, trembling taking place on the Moon. Rover has found indications for the presence of several minerals on the lunar surface. It has found evidence for the presence of elements like Aluminum, Iron, Calcium, Sulpher and Oxygen. Technologies developed for the Chandrayaan-3 mission can help develop and demonstrate new technologies required for interplanetary missions.

Know your Scientist



Dr. P. Veeramuthuvel was the project director of the Chandrayaan-3 programme. He was born on October 22, 1976, in Villupuram district and completed his schooling there. He finished his diploma in mechanical engineering, and then he got his master's degree. He obtained his doctorate degree from IIT, Madras. He joined ISRO in 2014, accomplishing his childhood dream of becoming a scientist at ISRO.

8.3 NASA (National Aeronautics and Space Administration)

NASA is the most popular space agency whose headquarters is located at Washington, USA. It was established on 1st October 1958. It has 10 field centers, which provide a major role in the execution of NASA's work. NASA is supporting International Space Station which is an international collaborative work on space research. It has landed rovers on Mars, analysed the atmosphere of Jupiter, explored Saturn and Mercury.

Mercury, Gemini and Apollo programs helped NASA learn more about flying in space. NASA's robotic space probes have visited every planet in the solar system. Satellites launched by NASA have revealed a wealth of data about Earth, resulting in valuable information such as a better understanding of weather patterns. NASA technology has contributed to make many items used in everyday life, from smoke detectors to medical tests.

8.3.1 Apollo Mission

Apollo missions are the most popular missions of NASA. These missions made American Astronauts land on the Moon. It consists of totally 17 missions. Among them Apollo-8 and Apollo-11 are more remarkable. Apollo-8 was the first manned mission to go to the Moon. It orbited around the Moon and came back to the Earth. Apollo-11 was the first 'Man Landing Mission' to the moon. It landed on the Moon on 20th July 1969. Neil Armstrong was the first man to walk on the surface of the Moon.

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